

Yurii Shchurko

AI Automation Engineer | n8n Specialist

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AI Automation Engineer specializing in autonomous agents and Advanced RAG architectures (pgvector, Pinecone, LLMs). Expert in n8n orchestration and custom JavaScript for Safe Data Mutation. Focused on automating workflows and optimizing operational costs via scalable AI solutions.

PROFESSIONAL EXPERIENCE

AI Automation Engineer (R&D) (02/2025 - 04/2026)

Personal AI assistant project Kyiv, Ukraine

- **Project Lifecycle:** Managed the end-to-end development of an n8n-based AI assistant from gathering client requirements and planning MVP features to technical R&D collaborating closely with a senior developer.
- **Architecture & integrations:** Architected a modular n8n and LangChain ecosystem. Built a Telegram-controlled Semantic Router to distribute tasks to specialized AI agents, integrating OpenAI and Notion via REST APIs.
- **Backend & NLP:** Developed a Node.js (Express.js) microservice utilizing NLP libraries (compromise) to parse natural language time data into dynamic Notion API filters.
- **Hybrid RAG search:** Engineered an advanced RAG engine (PostgreSQL pgvector, Pinecone) that combines full-text search, vector similarity, and strict SQL filtering for highly accurate retrieval.
- **ETL & data integrity:** Built automated n8n data pipelines using custom JavaScript. Implemented safe data mutation (deep-merge logic) for zero-loss Notion updates and serialized data for vectorization.
- **Workflow control:** Integrated Google Calendar and Telegram using a human-in-the-loop (two-phase commit) approach, requiring user validation for AI-generated schedules to eliminate LLM hallucinations.
- **Cost optimization:** Designed a PostgreSQL-backed LLM query planner to prevent unnecessary vector searches, significantly reducing API token costs.

Hospitality Administrator / Bar Manager (03/2020 - 12/2024)

- Managed daily operations, team schedules, and solved problems quickly under pressure. Built strong communication and software skills before moving into AI Automation.

EDUCATION

Master of Regional Development and Urban Planning (09/2018 - 06/2020)

Taras Shevchenko National University of Kyiv

SKILLS

- **Languages:** JavaScript (Node.js), SQL, express.js
- **AI & RAG:** LangChain, OpenAI, Pinecone, pgvector
- **Automation:** n8n, ETL Pipelines, REST APIs, Webhooks
- **Databases & DevOps:** PostgreSQL, Docker, Git
- **English:** B2
- **Ukrainian:** Native